

CHARACTERIZATION OF CRYSTAL EFFICIENCY: < *TORSION CORRECTION* >

*2024 dataset
for TCCP and TCCPA*

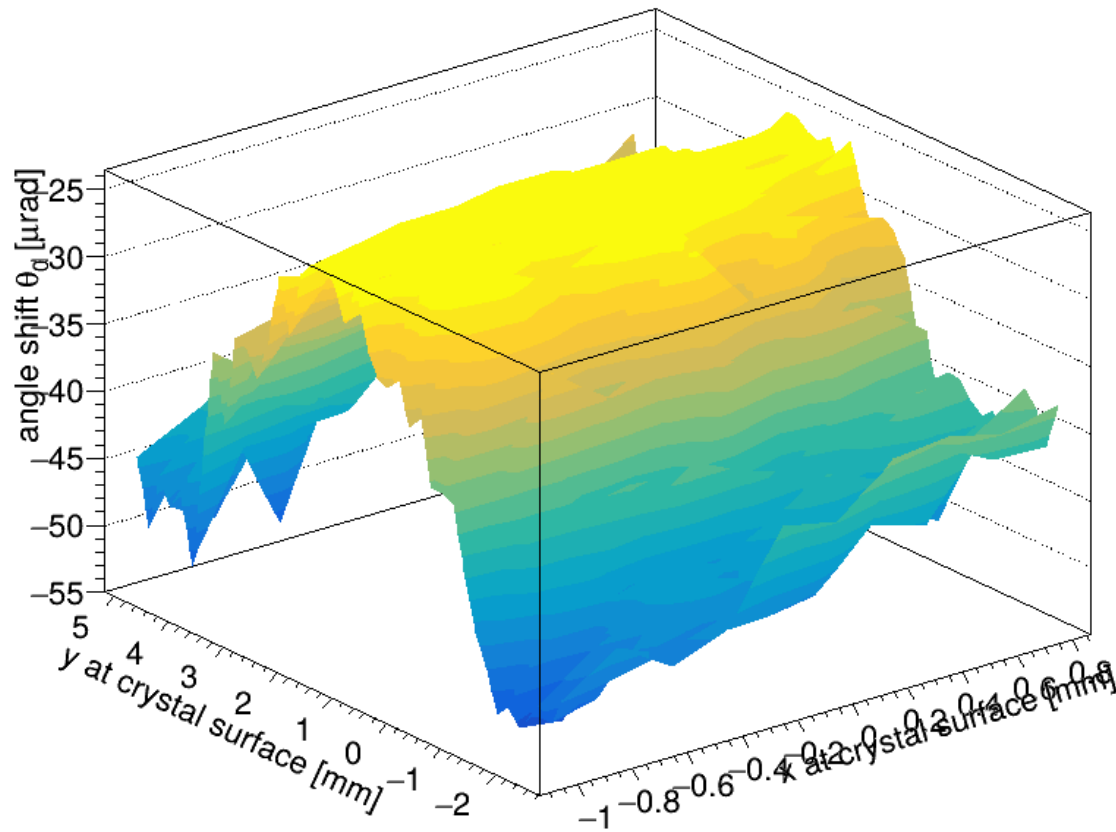
JULY 2026
ANNA SOFIA MORO

TORSION CORRECTION

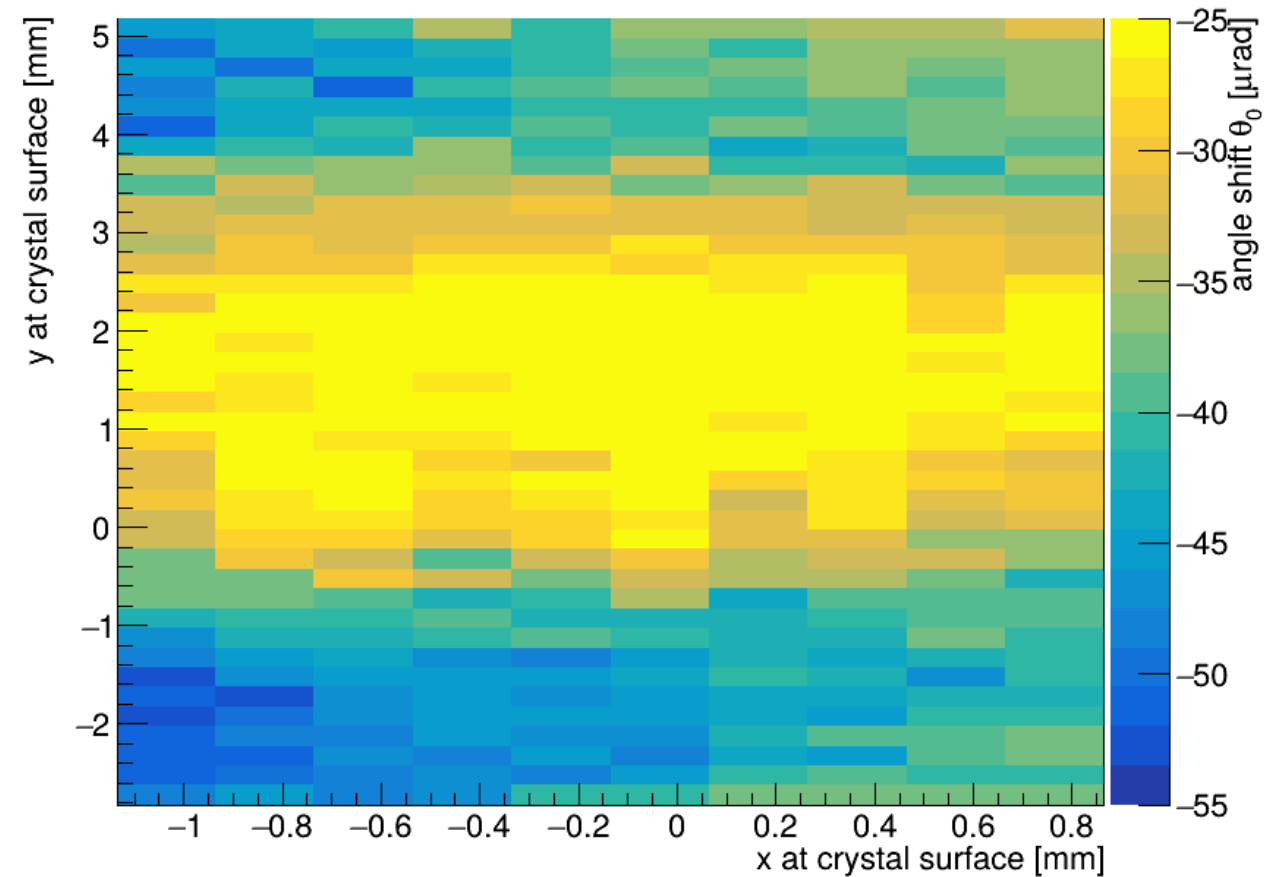
- For the longer crystals (TCCP and TCCPA), the effect of the torsion is not negligible and must be incorporated into the analysis.
- The effect of the torsion can be corrected by doing a checkerboard analysis:
 - By dividing the data sample into square bins (for example 0.1 mm x 0.1 mm) as a function of the impact position on the crystal surface
 - The angular shift parameter θ_0 is then varied to find the maximum channeling efficiency ϵ_{ch} for each bin
 - We obtain a 2D map of torsion variation across the crystal surface

TORSION CORRECTION

2D Torsion Map (bins 0.1 x 0.1 mm)



2D Torsion Map (bins 0.1 x 0.1 mm)



TORSION CORRECTION

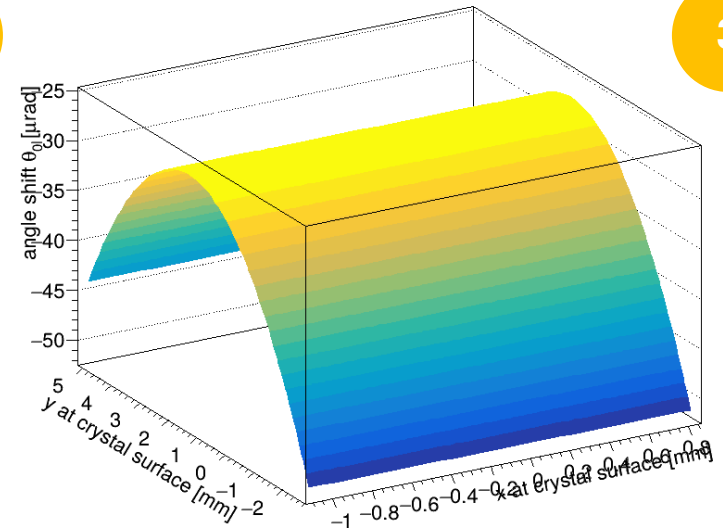
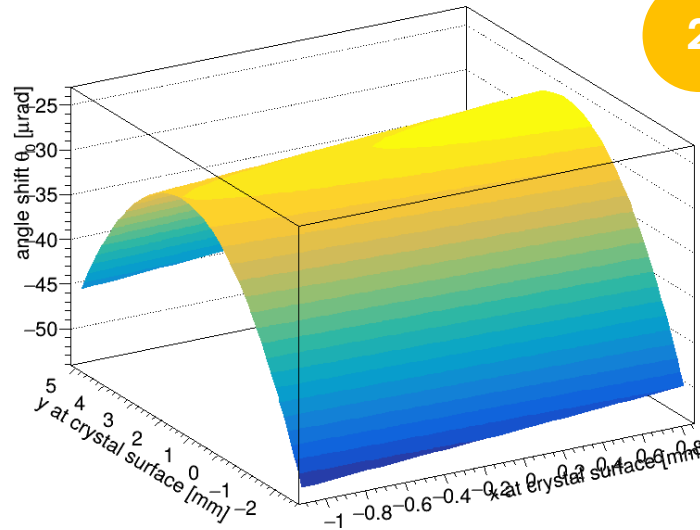
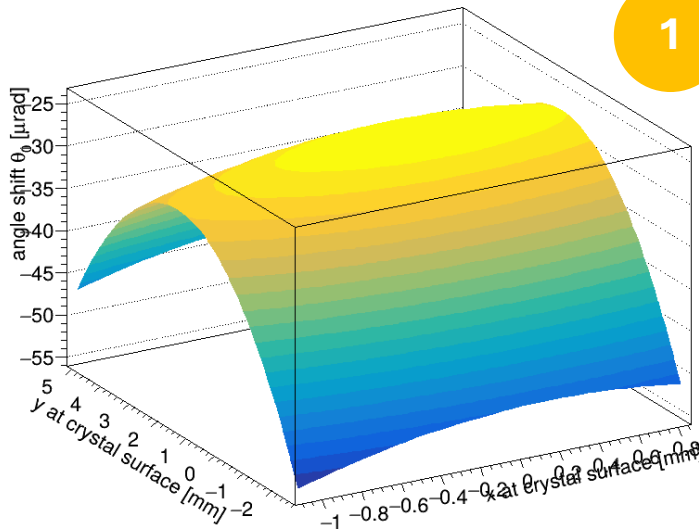
The 2d map is fitted through:

1. $\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 x^2 + p_4 y^2 + p_5 xy$
2. $\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 y^2$
3. $\theta_{corr}(x, y) = \theta_0 + \tau_y \cdot y + p_2 y^2$

Continuous 2D Torsion Map

Continuous 2D Torsion Map

Continuous 2D Torsion Map



TORSION CORRECTION

Now we should correct the effect of torsion using the parameter $\theta_0^{fit}(x, y)$ found for each bin.

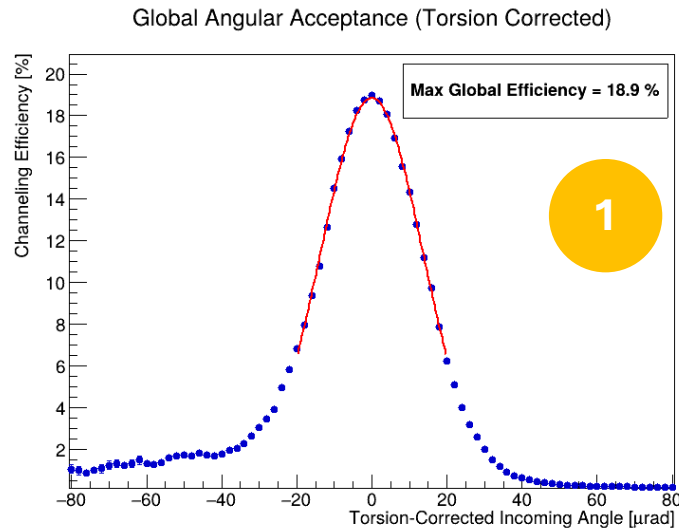
instead of the Lindhard cut $|\theta_{in,x}| \leq \frac{1}{2} \theta_L$ we will use

$$\left| \theta_x^{in} - \left(\theta_x^{off} + \tau_y \cdot d_y + \tau_x \cdot d_x \right) \right| \leq \frac{1}{2} \theta_L$$

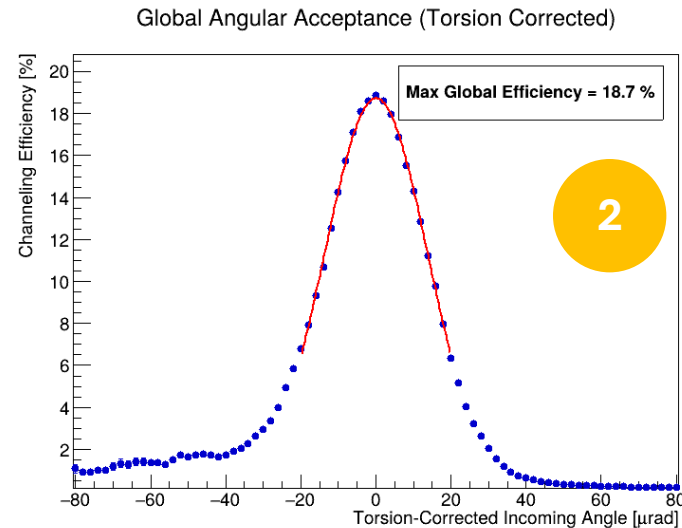
How am I behaving if the fit is not linear but parabolic?

EFFICIENCY COMPUTATION

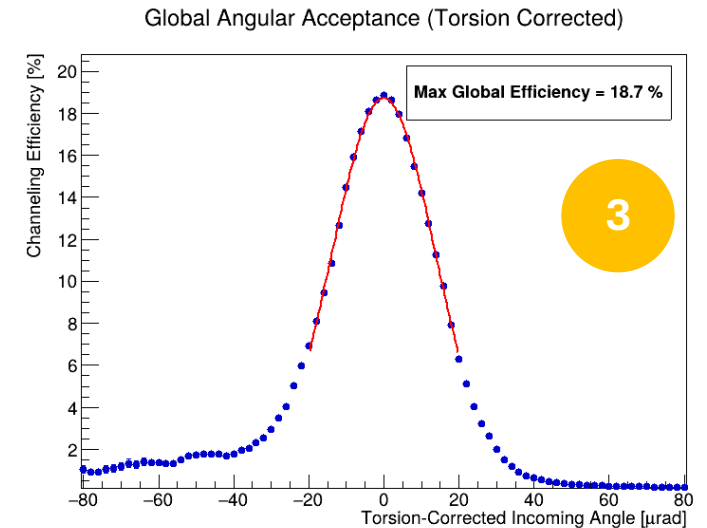
- If I keep the second-order corrections:



$$\tau_y = 4.11 \mu\text{rad/mm}$$
$$\epsilon_{ch} = 18.9\%$$



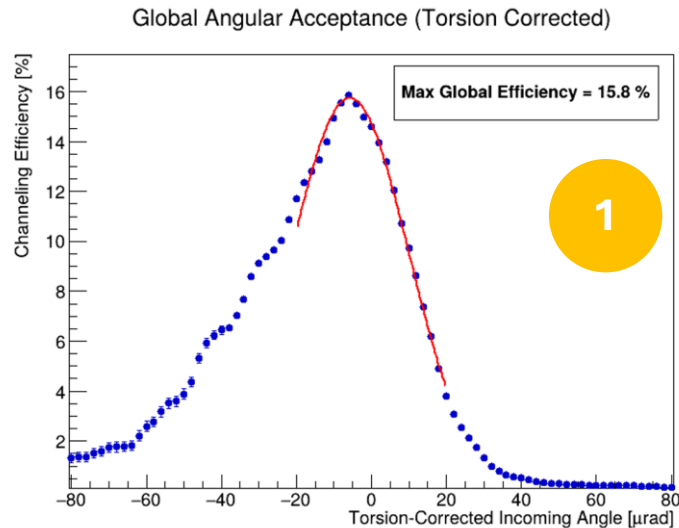
$$\tau_y = 4.13 \mu\text{rad/mm}$$
$$\epsilon_{ch} = 18.7\%$$



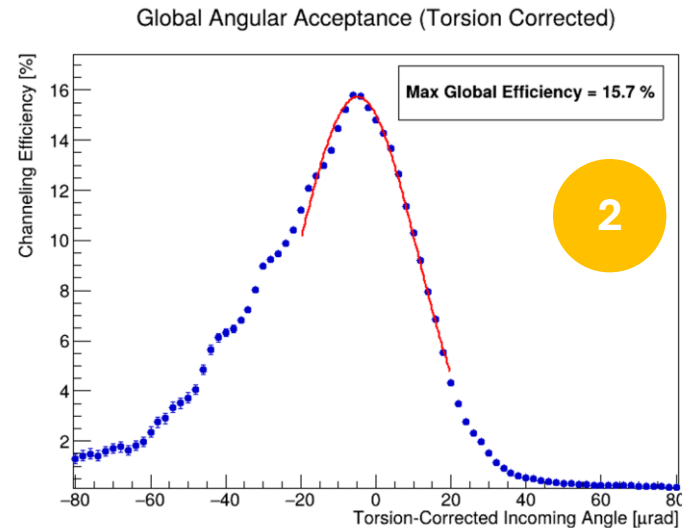
$$\tau_y = 4.11 \mu\text{rad/mm}$$
$$\epsilon_{ch} = 18.7\%$$

EFFICIENCY COMPUTATION

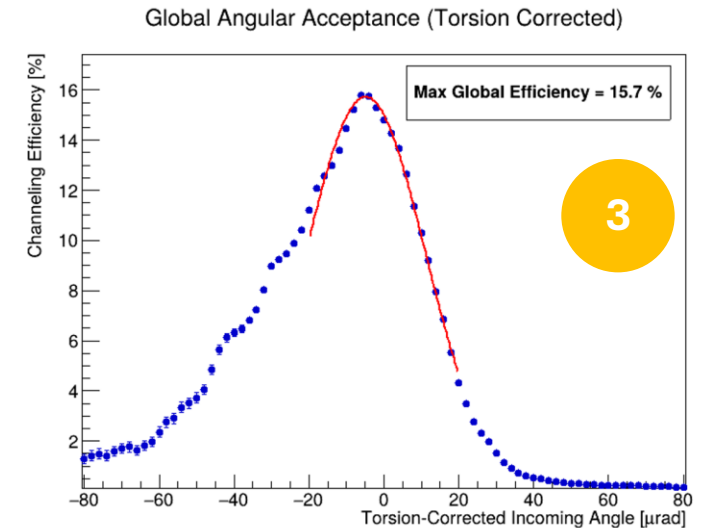
- If I do not (still doing a 3d fit):



$$\tau_y = 4.11 \mu\text{rad/mm}$$
$$\epsilon_{ch} = 15.8\%$$



$$\tau_y = 4.13 \mu\text{rad/mm}$$
$$\epsilon_{ch} = 15.7\%$$



$$\tau_y = 4.11 \mu\text{rad/mm}$$
$$\epsilon_{ch} = 15.7\%$$

TO DO